CGC University

Jhanjeri, Mohali

HACXLERATE 2026 | VIRTUAL ROUND 1

Powered by Kampus.VC | Digital Risk Protection challenge

Rakshak

Digital Risk Protection for Social & App Monitoring. Find fake profiles, scam accounts and impersonating apps before your customers do.

TEAM OPENCODERS

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Brands are being impersonated

Attackers copy a brand's name, logo and tone to deceive customers, steal credentials and spread scams.

Fake profiles

Copy name, logo, branding

Fake brand pages

Pose as the official presence

Scam accounts

Giveaways and refund traps

Fake apps

Imitate the real app

THE CHALLENGE

Build a Digital Risk Protection platform for social and app monitoring that finds impersonation and ranks what to act on first.

Real or fake? Spot the difference

AP

ApexPay

@apexpay

OFFICIAL

The official handle for ApexPay Inc. Modern financial infrastructure.

Registered asset, never flagged

AP

ApexPay Support

@apexpay_support_desk

UNVERIFIED

Official 24/7 support. DM to unlock frozen balances or claim bonuses: bit.ly/apex-verify

Joined 3 days ago

Look-alike name | Scam wording | Off-platform link | New account

Illustrative example, not a real account

Why it hurts

- Customers deceived
- Credentials stolen
- Brand blamed for scams
- Users sent to fraud

Meet Rakshak, your brand protector

01



Define

Register the brand and its official assets

02



Collect

Search app stores, import or submit suspects

03



Detect

Exclude official assets, compare, score 0-100

04

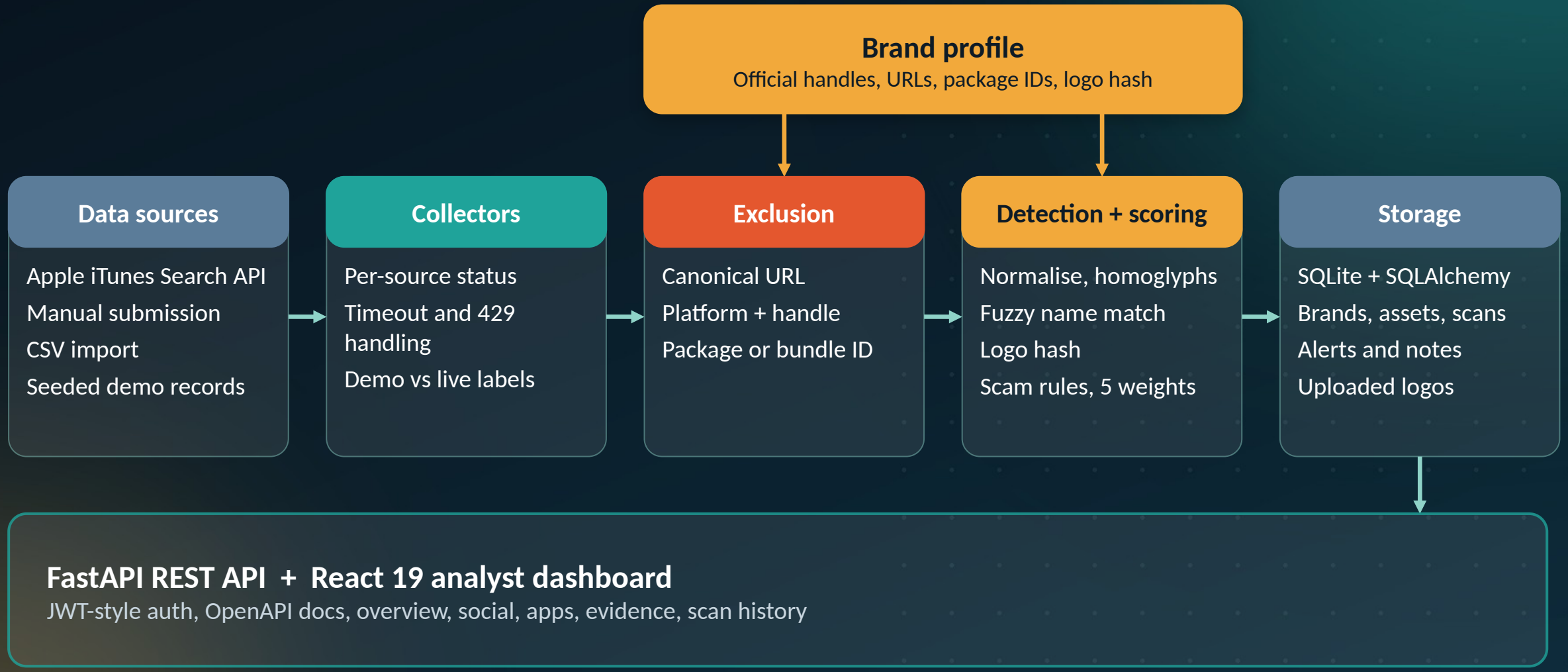


Act

Review evidence, triage, request takedown

Which accounts and apps are pretending to be us, and what should we act on first?

System architecture



One profile defines what's official

Brand profile (example)

BRAND NAME

ApexPay

ALIASES

Apex Pay, APX

OFFICIAL LOGO

Uploaded and hashed

OFFICIAL ACCOUNT

@apexpay (twitter)

OFFICIAL APP

com.apexpay.mobile

Exclusion registry

- ✓ Canonical URL match
- ✓ Platform + handle match
- ✓ Package or bundle ID match

↓

Official asset: excluded, never scored

Display names are never used for exclusion.

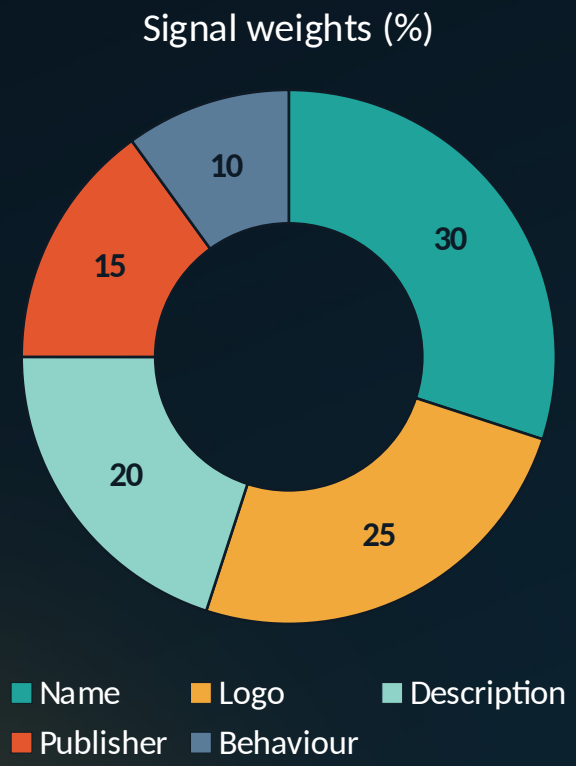
Catching look-alike names

REAL BRAND **ApexPay**

ApexPay	Homoglyph	Cyrillic e looks identical; NFKC and a confusable map expose it
Ap3xPay	Digit swap	Leetspeak digits are mapped back to letters
AppexPay	Typo	Edit distance and transposition checks
ApexPay Official Support	Added words	Brand plus risk words such as support, verify, giveaway
apex_pay	Delimiter	Punctuation stripped, then flagged as evasion

Guardrail: a name match alone never reaches Critical

Explainable risk scoring



ILLUSTRATIVE ALERT

78

HIGH

Confidence: medium

Name match

Scam wording

Publisher

Missing evidence lowers confidence

Name-only match never reaches Critical

Every alert explains why it was flagged

Low alert from 40

Medium 45+

High 65+

Critical 80+

Social media monitoring

Names and handles

Homoglyphs, typos, added words

Logo mimicry

Perceptual hash match

Scam wording

Refund, giveaway, verify

Off-platform links

Telegram, WhatsApp, shorteners

New, unverified

No link to the real brand

How candidates enter

Manual submission

Analyst reports a suspect profile

CSV import

Up to 5 MB and 5,000 rows

Demo records

Labelled synthetic examples

App store monitoring

LIVE

Apple App Store

Public iTunes Search API. No key needed. Outages and rate limits handled.

DEMO

Seeded app records

Clearly labelled, never shown as live results.

Title similarity

Same look-alike engine as social names

Publisher mismatch

Checked against the registered publisher

Description overlap

Wording that resembles the real app

Scam wording

Direct APK, loans, giveaways

Official app exclusion

Matched by bundle or package ID

Built for analysts

Every source reports its health

completed | partial | unavailable | rate_limited | timeout | failed

A source outage never stops the scan, and never produces invented results.

Triage workflow



Setup wizard
Guided brand registration

Evidence view
Per-signal breakdown

Honest labels
Demo and live never mixed

Scan history
Audit of every scan

Built to score on every criterion

30%

Social monitoring

Names, logos, scam wording, bio links. Official handles never flagged.

30%

App store monitoring

Live Apple search, publisher mismatch, description overlap.

15%

Brand profile

Validated official assets that drive exclusion and scoring.

15%

User experience

Setup wizard, clear evidence, source status, triage.

10%

Demo and slides

Working prototype, architecture, demo script, this deck.

BONUS **Look-alike detection: homoglyphs, digit swaps, typos, added words and delimiters**

The tech stack



Backend

Python / FastAPI / SQLAlchemy / SQLite / Pydantic



Detection

RapidFuzz / Unicode NFKC / imagehash / Pillow



Frontend

React 19 / TypeScript / Vite / Tailwind / Recharts



Security

PBKDF2 / signed tokens / per-user data scoping



DevOps

Docker Compose / Nginx / health checks



Testing

pytest / FastAPI TestClient / Mermaid docs

Secure, tested, deployable



Secure

- Salted PBKDF2 passwords
- Signed, expiring tokens
- Per-user data isolation
- Upload checks, strict CORS



Tested

- Similarity and exclusion
- Scoring guardrails
- Outages and rate limits
- Auth isolation, end-to-end flow



Deployable

- Docker Compose + Nginx
- Health checks
- Persistent data volume
- Non-root API container

What's next

1 Live social intake

Platform APIs behind the collector interface

2 Google Play

Extend app coverage beyond Apple

3 Live icon matching

Hash icons from live app results

4 Stricter name checks

Benign-name list, whole-word keywords

5 Image embeddings

Catch recoloured or cropped logos

6 Scale and respond

PostgreSQL, scheduled scans, takedowns

Thank you

Questions?

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